# RNA-Seq Literature Familiarization and Data Characterization

**Student: _Nesie D. Calipas** **Group #: 1** **Assigned Topic: Heat Stress**

## Part 3. Paper Information (Group Answer)

| Item | Answer |
|---|---|
| Title of the paper | Transcriptomic profiling of Paulownia elongata in response to heat stress |
| Authors | Neerja Katiyar, Niveditha Ramadoss, Dinesh Gupta, Suman B. Pakala, Kerry Cooper, Chhandak Basu |
| Year | 2021 |
| Journal | Plant Gene |
| Organism | Paulownia elongata |
| Tissue / organ / cell type | Leaves |
| Biological question | How does P. elongata respond at the transcriptomic level to heat stress? |
| Control condition | 25°C |
| Treatment or stress condition | 40°C for 24 hours |
| Number of biological replicates | 3 control + 3 heat-stress replicates |
| Sequencing platform | Illumina MiSeq |
| Single-end or paired-end sequencing | Paired-end |
| Read length, if reported | 76 bp |
| RNA-seq repository | NCBI Sequence Read Archive (SRA) |
| BioProject / Study accession | PRJNA485845 |
| Run accession numbers | SRP158908 (SRA study; individual SRR run numbers not given in the paper) |
| Reference genome or transcriptome used by the authors | De novo P. elongata transcriptome assembled with Trinity v2.0.6 |

| Item | Answer |
|---|---|
| Main RNA-seq software or tools reported by the authors | Trinity, DIAMOND, MEGAN6, OmicsBox/BUSCO, Blast2GO, Salmon, edgeR |

## Part 4. Experimental Design Diagram

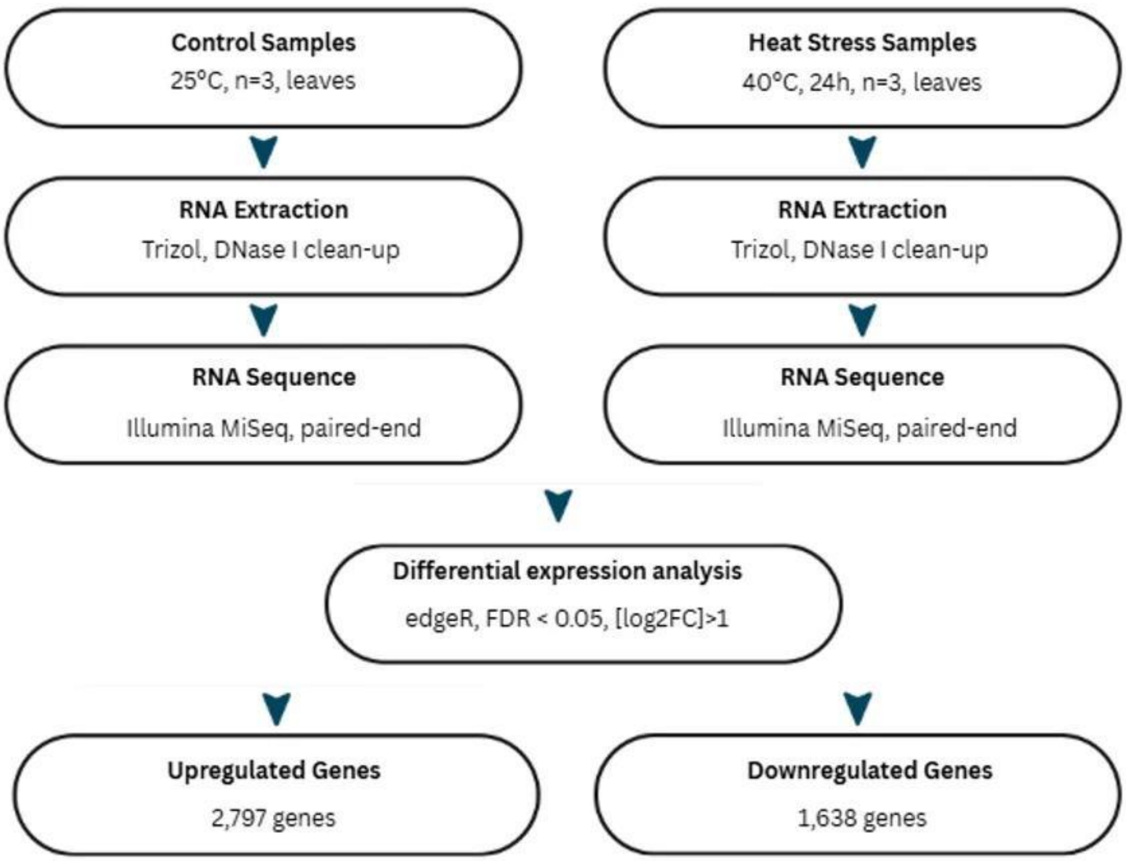


Figure 1. Experimental design based on the Transcriptomic Profiling of Paulownia elongata in Response to Heat Stress paper.

• What was changed or manipulated?

- Temperature (40°C for 24 hours vs. 25°C control)

• What was the control?

- Plants grown at 25°C, same photoperiod

• What was measured?

- Genome-wide gene expression (RNA-seq → differential expression)

• How many biological replicates were used?

- 3 per condition (H1–H3 heat stress, C1–C3 control)

• What tissue or cell type was sequenced?

- Leaves

## Part 6. My Assigned RNA-Seq Sample

| Item | Answer |
|---|---|
| Run accession number | SRR7758330 |
| Condition (control/treatment) | Heat-treated |
| Biological replicate number, if known | Sample 3 |
| Single-end or paired-end | Paired-end (R1 + R2) |
| Approximate file size, if available | 192 MB |

## Part 8. FASTQ File Examination

1. How is a FASTQ file different from the FASTA genome file used in the previous assignments?

   - A FASTQ file contains the actual sequencing reads along with their quality scores, while the FASTA genome file contains the DNA sequences without quality scores. In simple terms, FASTA shows the sequence itself, while FASTQ shows both the sequence and how reliable each base was during sequencing.

2. What information is present in FASTQ that is not present in FASTA?

   - FASTQ contains quality scores for each base in the sequencing read, which show how confident the sequencing machine is that each nucleotide was read correctly. FASTA only provides the DNA sequence and does not include these quality scores.

3. FASTQ contains a quality score for each base in every read. This indicates how confident the sequencing machine is that each nucleotide was read correctly. FASTA does not contain these quality scores.?

- Quality scores are important because they help identify low-quality or unreliable bases in the sequencing reads. This helps ensure that the reads used for further analysis, such as alignment and gene expression analysis, are accurate and reliable.

4. What information is present in FASTQ that is not present in FASTA?

   - FASTQ contains quality scores for each nucleotide/base in every read. These scores indicate how confident the sequencing machine is that each base was read correctly. FASTA does not contain this quality information.

5. Why are quality scores important in RNA-seq?

   - Quality scores are important because they help determine whether the sequencing reads are reliable or contain low-quality bases. Low-quality reads can affect read alignment and gene expression results, so checking them helps ensure more accurate RNA-seq analysis.

## Part 10. My RNA-Seq Dataset Characterization (from FastQC/Galaxy)

| **Item** | **Answer** |
|---|---|
| RNA-seq run accession | SRR7758330 |
| Condition | Control |
| Single-end or paired-end | Paired-end |
| Number of reads / sequences | 3,441,734 reads |
| Read length | 35–76 bp |
| File size | 192 MB |
| GC content | 41% (R2); 42% (R1) |
| General per-base sequence quality | Pass |
| Presence of adapter content | None detected / Pass |
| Presence of overrepresented sequences | None detected / Pass |
| Any quality warning observed | Yes — Sequence Length Distribution warning; Per Base Sequence Content failed |

## Part 12. Group Comparison Table

| Student | Accession | Condition | Reads | Read length | GC % | Main QC observation |
|---|---|---|---|---|---|---|
| Calipas | SRR7758330 | Heat-treated | 3,441,734 | 35–76 bp | 41% | Overall good quality; per-base sequence content failed and sequence length distribution warned |
| Bullanday | SRR7758328 | | 4,956,271 | 35–76 bp | 45% | Per base sequence content failed |
| Alama | SRR7758331 | | 3.147,060 | 35–76 bp | 39% | R1 failed "Per Base Sequence Content"; mild duplication/length-distribution warnings; and trace polyA adapter content |
| Ferrer | SRR7758332 | | 1,349,099 | 35–76 bp | 41% | Per base seq. content FAILED; Seq. Length WARNING |

## Part 13. Interpretation Questions

1. What biological question was the original RNA-seq study trying to answer?

   Which genes and biological pathways in Paulownia elongata change expression in response to heat stress (40 °C), helping to explain its heat tolerance?

2. Why did the authors use RNA-seq instead of only examining the genome?

   RNA-seq was used because the genome sequence itself does not change with temperature. Only by measuring RNA levels can researchers see which genes are turned up or down in response to heat.

3. What is the difference between genomic DNA and the RNA molecules measured by RNA-seq?

   Genomic DNA is the stable genetic material present in essentially every cell. The RNA molecules quantified by RNA-seq are temporary transcripts whose abundance reflects current gene activity and can change rapidly with environmental conditions.

4. What is a biological replicate and why is it important?

   A biological replicate is an independent biological sample (for example, a different plant) that received the identical treatment. Replicates are essential for distinguishing true biological differences from random variation.

5. What is a biological replicate and why is it important?

   In single-end sequencing only one end of each fragment is read. In paired-end sequencing both ends are read, providing higher mapping accuracy, better detection of splice junctions, and improved resolution of repetitive regions.

6. What is the difference between single-end and paired-end sequencing?

   A FASTQ file is a plain-text format that stores raw sequencing reads. Each read occupies four lines: a header, the nucleotide sequence, a separator line, and a corresponding string of quality scores

7. What is a FASTQ file?

   FastQC evaluates the technical quality of raw sequencing data. It reports total read number, read length, GC content, per-base quality, adapter content, over-represented sequences, and other diagnostics that indicate whether the data are suitable for downstream analysis.

8. What information does FastQC provide?

   A high per-base quality score means the sequencer assigned that base with high confidence (low probability of error). Scores in the mid-to-high 30s are considered excellent.

9. What does a high per-base quality score indicate?

   Adapter sequences are artificial and not part of the biological sample. If retained, they can cause misalignment, reduced mapping rates, and inaccurate quantification of gene expression.

10. Why can adapter contamination be a problem?

    Overall, the samples in the group were of comparable quality. Read lengths and GC contents were similar, and the same mild QC pattern (mainly per-base sequence content bias) appeared across samples. Read counts varied but remained within an acceptable range.

11. Were all RNA-seq samples in your group similar in quality? Explain.

    Yes. Multiple samples, including the heat-treated sample SRR7758328, showed a failure in the Per base sequence content module. This is a frequent and generally non-critical feature of RNA-seq data caused by random-hexamer priming bias.

12. Did any sample show a possible quality problem? What was it?

Before differential expression can be performed, the reads must be quality-trimmed and adapter-cleaned, aligned or quasi-mapped to a reference transcriptome, quantified, normalized, and then statistically compared between control and treatment groups (for example with edgeR or DESeq2).

13. What additional steps would be needed before the researchers could compare gene expression between control and treatment samples?

## Part 15. Group Summary (one per group)

1. What study did the group choose and why is it relevant to the assigned topic?

- The group selected Katiyar et al. (2021), "Transcriptomic Profiling of Paulownia elongata in Response to Heat Stress," because it directly addresses the group's assigned topic of heat stress by identifying the genes and pathways behind P. elongata's heat tolerance using RNA-seq.

2. What organism, tissue, control, and treatment were used?

- P. elongata leaves; control plants grown at 25°C; treatment plants exposed to 40°C for 24 hours; 3 biological replicates per condition.

3. How many RNA-seq samples were included in the original study?

- 6 total — 3 control (C1–C3) and 3 heat-stressed (H1–H3)

.4. What sequencing platform and read type were used?

- Illumina MiSeq (MiSeq Reagent Kit v3), paired-end, 76 bp reads.

6. What did the group learn from examining the raw RNA-seq data?

- The FASTQ files followed the expected four-line structure with per-base Phred quality scores; overall base quality was high (mostly Phred 34–38) across the group's samples, but read depth differed substantially between individually assigned runs, and a recurring "Per Base Sequence Content" failure showed up specifically in the two control replicates.

6. Were there noticeable differences in data quality among samples?

- Yes. Read counts and GC% varied across the four runs, and the two controlcondition samples shared the same QC failure pattern, distinct from the treatment samples.

7. What analysis did the original authors perform after quality control?

- The authors built a de novo transcriptome with Trinity (no reference genome existed), screened for contamination with DIAMOND/BLAST and MEGAN6, assessed assembly completeness with BUSCO, annotated transcripts withBlast2GO/OmicsBox, quantified expression with Salmon, and identified differentially expressed genes with edgeR (FDR < 0.05, |log2FC| > 1), followed by GO and KEGG pathway enrichment.

8. What will the group need to learn next in order to reproduce part of the authors' gene-expression analysis?

- Read trimming/adapter removal, alignment or pseudo-alignment of reads to a reference or assembled transcriptome, transcript quantification (e.g., Salmon), and differential expression testing (e.g., edgeR), the same pipeline steps the original authors used to go from raw reads to their list of upregulated/downregulated genes.

**Github link:**

https://github.com/rhealynalama-bio/RNASeq_Group-1_Heat-Stress